

Rethinking case attraction on Ancient Greek infinitive clauses

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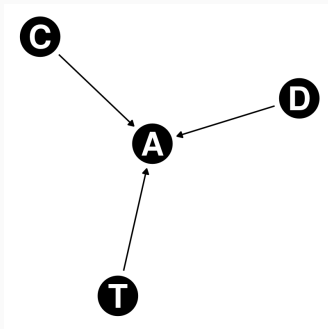


- (1) a. συμβουλεύει **τῷ Ξενοφῶντι ἐλθόντα** εἰς Δελφοὺς
ἀνακοινῶσαι τῷ θεῷ περὶ τῆς πορείας.
He advises Xenophon to go to Delphi and ask the god
about the travel. (Xen. *Anab.* 3.1.5)
- b. ἀφῆκε **μοι ἐλθόντι** πρὸς ὑμᾶς λέγειν τάληθῃ.
He allowed me to go and speak the truth in front of you.
(Xen. *Hell.* 6.1.13)

Direct Acyclic Graph: Agreement

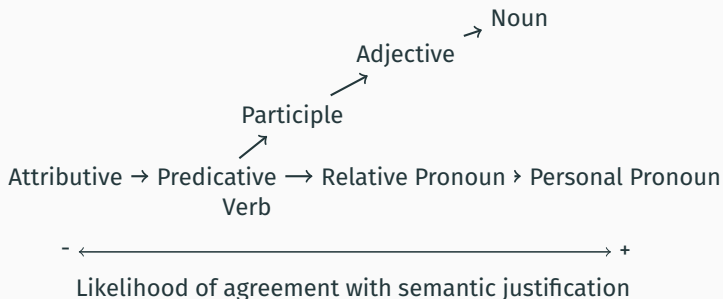
Framework: Corbett (2006)

Features of the **C**ontroller, **T**arget and **D**omain “cause”
non-canonical **A**greement

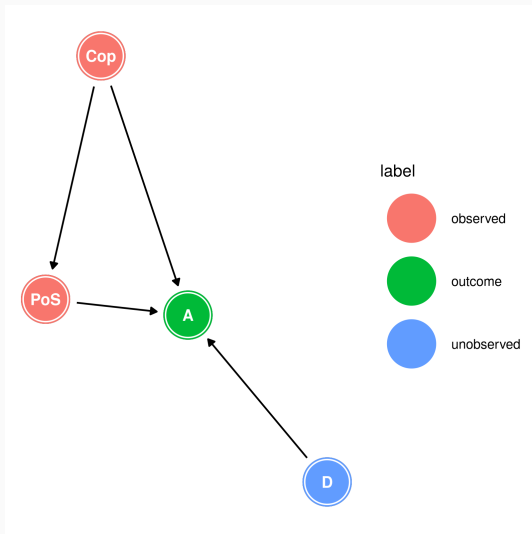


Domain of attraction: Part of speech and copular infinitive

(2) Agreement and Predicate Hierarchies (Corbett 2006, p. 233):

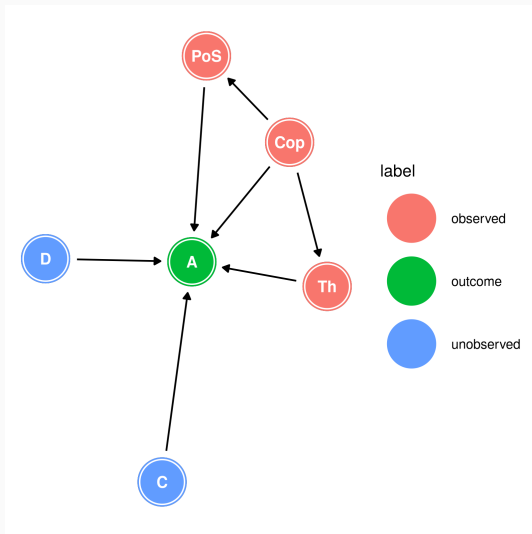


The domain of attraction: DAG

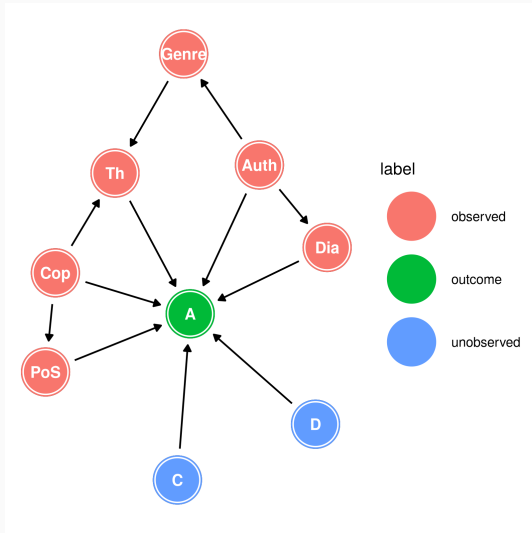


- (3) ἀλλ' ἐξαρκῇ **σοι** τὰ ἔργα αὐτῶν **ὀρῶντι** σέβεσθαι καὶ τιμᾶν τοὺς θεοὺς.
But it is enough for you to look at these deeds and respect the gods. (Xen. *Mem.* 4.3.13)

The controller of attraction: DAG



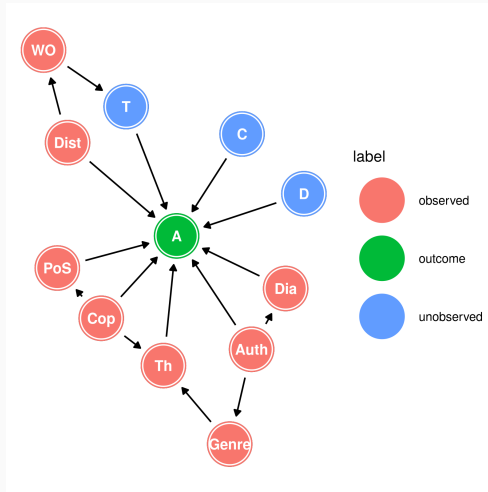
Issues of dialect and authorship: DAG



The target of attraction: distance and precedence

- (4) ὥς **ἀγαθοῖς** τε **ὑμῖν** προσήκει εἶναι.
That you may be good people. (Xen. *Anab.* 3.2.11)

The target of attraction: final DAG



Tooling: R (R Core Team 2013), the packages `tidyverse` (Wickham et al. 2019), `dagitty` (Textor et al. 2016), `ggdag` (Barrett 2025), and `rethinking` (McElreath 2020), as well as the software Stan (Stan Development Team 2025a,b).

Generalized linear model

$$A \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha + \beta_{\text{Cop}} \text{Cop}_i + \beta_{\text{Dist}} \text{Dist}_i + \beta_{\text{Wo}} \text{Prec}_i$$

$$+ \beta_{\text{PoS}} \sum_{j=0}^{\text{PoS}_i-1} \delta_{j\text{PoS}} + \beta_{\text{Th}} \sum_{j=0}^{\text{Th}_i-1} \delta_{j\text{Th}}$$

$$+ \beta_{\text{Auth}_i} + \beta_{\text{Dia}_i}$$

$$\alpha \sim \text{Normal}(0, 1)$$

$$\beta_{\text{Cop}}, \beta_{\text{Dist}}, \beta_{\text{Prec}} \sim \text{Normal}(0, 1)$$

$$\beta_{\text{PoS}}, \beta_{\text{Th}} \sim \text{LogNormal}(0, 1)$$

$$\delta_{\text{PoS}}, \delta_{\text{Th}} \sim \text{Dirichlet}(2)$$

$$\beta_{\text{Auth}} = \mathbf{z}_{\text{Auth}} * \sigma_{\text{Auth}}$$

$$\beta_{\text{Dia}} = \bar{\beta}_{\text{Dia}} + \mathbf{z}_{\text{Dia}} * \sigma_{\text{Dia}}$$

$$\bar{\beta}_{\text{Dia}}, \mathbf{z}_{\text{Auth}}, \mathbf{z}_{\text{Dia}} \sim \text{Normal}(0, 1)$$

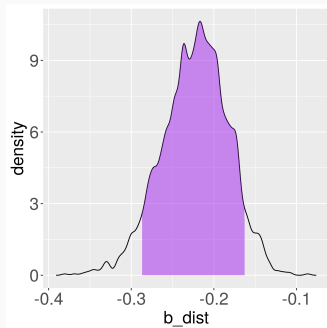
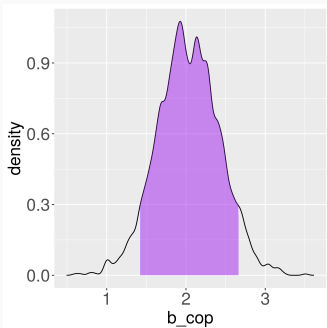
$$\sigma_{\text{Auth}}, \sigma_{\text{Dia}} \sim \text{Exponential}(1)$$

A word on Ordered Categorical Predictors

PoS	Cummulative effect
Participle	$\beta_{\text{PoS}} * 0$
Adjective	$\beta_{\text{PoS}} * (\delta_{\text{Adj}})$
Noun	$\beta_{\text{PoS}} * (\delta_{\text{Adj}} + \delta_{\text{Noun}})$

Copula and Distance

Effect	mean	sd	5.5%	94.5%	rhat
β_{Copula}	2.03	0.39	1.42	2.66	1
β_{Dist}	-0.22	0.04	-0.29	-0.16	1



Part of Speech and the Predicate Hierarchy

Effect	mean	sd	5.5%	94.5%	rhat
β_{PoS}	0.45	0.28	0.11	0.95	1
δ_{Adj}	0.53	0.18	0.23	0.82	1
δ_{Noun}	0.47	0.18	0.18	0.77	1

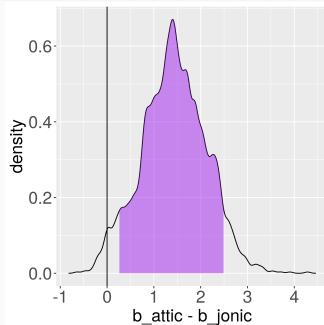
PoS	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Participle	$\beta_{\text{PoS}} * 0$	0.00	0.00	0.00	0.00	NA
Adjective	$\beta_{\text{PoS}} * \delta_{\text{Adj}}$	0.25	0.18	0.05	0.60	1
Noun	$\beta_{\text{PoS}} * (\delta_{\text{Adj}} + \delta_{\text{Noun}})$	0.45	0.28	0.11	0.95	1

Thematic Role of the Controller

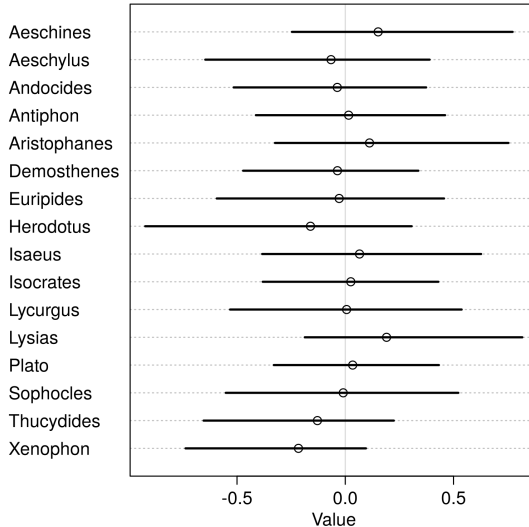
Effect	mean	sd	5.5%	94.5%	rhat
β_{Theta}	2.47	0.40	1.84	3.12	1
δ_{Exp}	0.34	0.11	0.17	0.51	1
δ_{Agent}	0.66	0.11	0.49	0.83	1

Theta	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Recipient	$\beta_{\text{Th}} * 0$	0.00	0.0	0.00	0.00	NA
Experiencer	$\beta_{\text{Th}} * \delta_{\text{Exp}}$	0.83	0.3	0.38	1.34	1
Agent	$\beta_{\text{Th}} * (\delta_{\text{Exp}} + \delta_{\text{Agent}})$	2.47	0.4	1.84	3.12	1

Effect	mean	sd	5.5%	94.5%
$\bar{\beta}_{\text{Dia}}$	-0.49	0.84	-1.78	0.90
σ_{Dia}	1.20	0.78	0.30	2.67
β_{Attic}	-0.16	0.89	-1.55	1.28
β_{Jonic}	-1.59	1.01	-3.19	-0.02
$\beta_{\text{Attic}} - \beta_{\text{Jonic}}$	1.43	0.69	0.26	2.49



Authorship



Thank you!

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Data and code available at

`github.com/caiogeraldes/icag12025`